A Survey of Radiomics and Deep Learning for Tumor Diagnosis and Prognosis

InteractiveSurvey

Abstract

The rapid evolution of medical imaging has transformed oncology by enabling the extraction of high-dimensional data that correlates with genomic profiles and treatment responses. Consequently, integrating advanced computational algorithms with imaging has shifted clinical paradigms from qualitative assessment to data-driven decision support. This survey comprehensively examines the synergistic application of handcrafted radiomics and deep learning methodologies for tumor diagnosis and prognosis, evaluating their respective strengths in feature engineering and model architecture. By synthesizing current literature, this work clarifies optimal deployment scenarios for specific algorithmic frameworks to guide the development of robust, clinically actionable biomarkers. The review details how classical pipelines, including statistical texture descriptors and topological data analysis, offer indispensable interpretability in data-scarce environments, while deep learning automates feature discovery. Furthermore, it critically assesses hybrid approaches that fuse handcrafted and learned representations, addressing critical challenges regarding reproducibility, standardization, and the translation of models into real-world clinical settings. Ultimately, this holistic synthesis bridges the gap between theoretical algorithm development and clinical implementation, providing a structured roadmap for selecting methodologies based on data availability and clinical objectives. By mapping the interdependencies between feature engineering paradigms and classification frameworks, this paper serves as an essential reference for developing next-generation imaging biomarkers that are both statistically robust and clinically viable for improving patient outcomes.

1 Introduction

The rapid evolution of medical imaging technologies has revolutionized oncology by providing non-invasive windows into tumor biology, enabling precise characterization of disease heterogeneity beyond visual inspection. As quantitative imaging biomarkers gain traction, the extraction of high-dimensional data from standard modalities like CT, MRI, and PET has become central to modern precision medicine. These data points capture subtle textural, morphological, and intensity variations that often correlate with underlying genomic profiles, treatment responses, and patient survival outcomes. Consequently, the integration of advanced computational algorithms with medical imaging has shifted the paradigm from qualitative assessment to data-driven decision support, promising to enhance diagnostic accuracy and prognostic stratification in clinical workflows.

This survey paper comprehensively examines the synergistic application of handcrafted radiomics and deep learning methodologies for tumor diagnosis and prognosis [1]. While deep learning offers automatic feature extraction capabilities, classical radiomics pipelines remain indispensable for their interpretability and efficacy in data-scarce environments [2]. The primary objective of this review is to dissect the methodological foundations of both approaches, evaluating their respective strengths in feature engineering, model architecture, and clinical validation. By synthesizing current literature, this work aims to clarify the optimal scenarios for deploying specific algorithmic frameworks, ultimately guiding researchers and clinicians toward robust, reproducible, and clinically actionable imaging biomarkers.

The initial portion of this review delineates the core paradigms of handcrafted radiomics, focusing on the sophisticated engineering required to transform raw images into meaningful quantita-

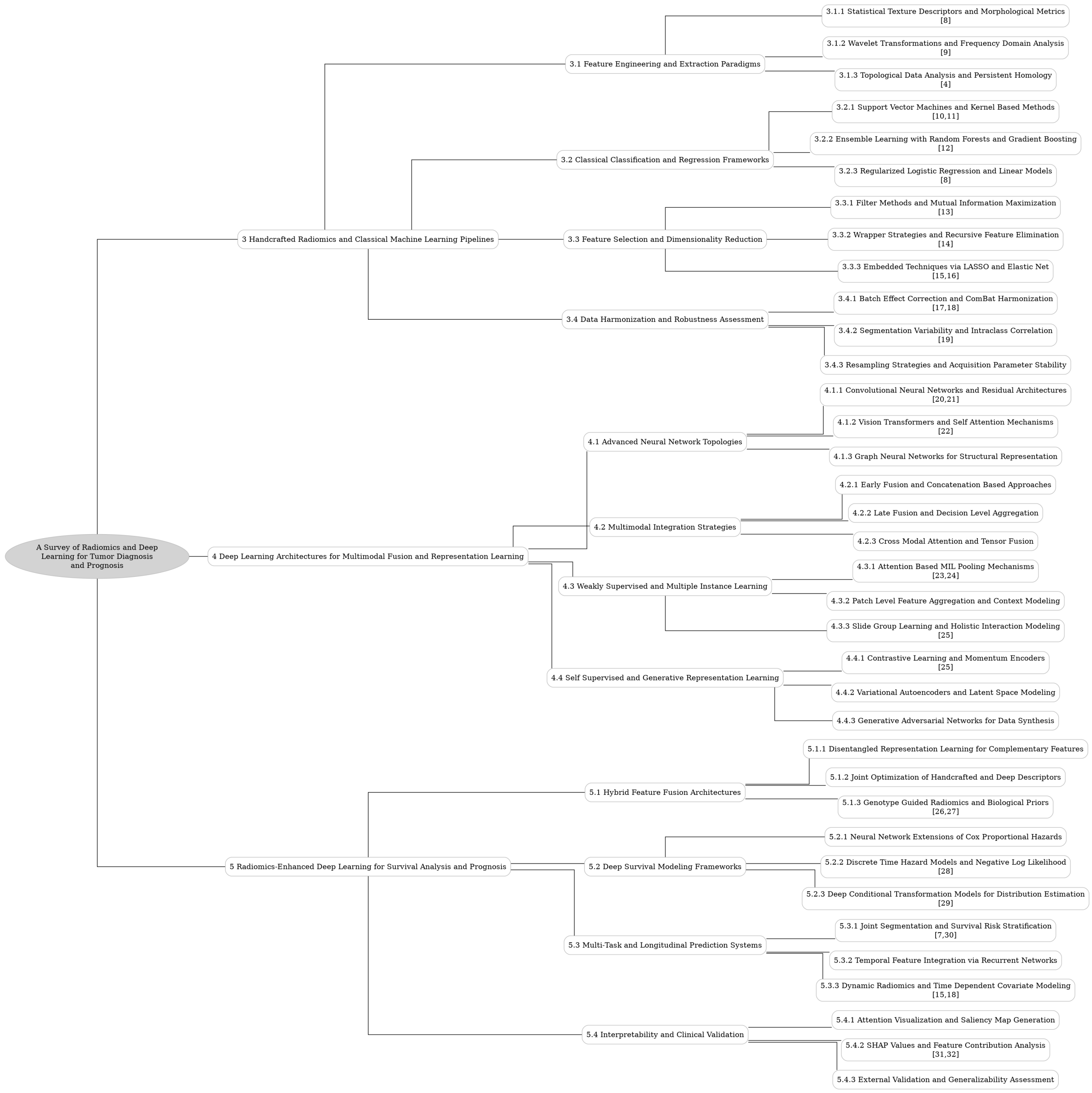

Figure 1: The outline of our survey: A Survey of Radiomics and Deep Learning for Tumor Diagnosis and Prognosis

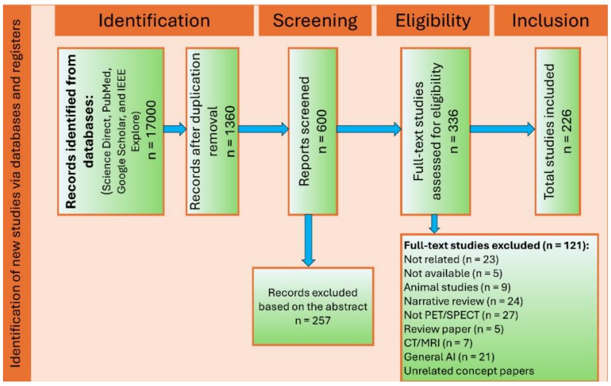

Figure 2: Chart from 'Handcrafted vs. Deep Radiomics vs. Fusion vs. Deep Learning: A Comprehensive Review of Machine Learning -Based Cancer Outcome Prediction in PET and SPECT Imaging' (arXiv:2507.16065)

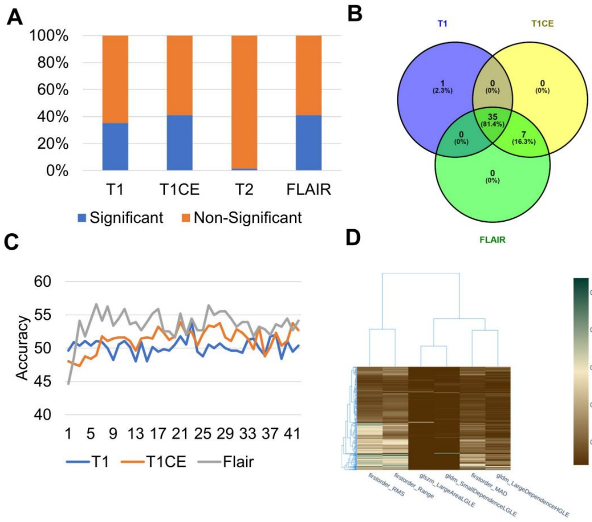

Figure 3: Chart from 'Unmasking unlearnable models: a classification challenge for biomedical images without visible cues' (arXiv:2407.19773)

tive descriptors [3]. It explores statistical texture descriptors and morphological metrics that quantify tissue heterogeneity and geometric complexity, alongside the critical role of wavelet transformations in isolating multi-scale frequency components. Furthermore, the discussion extends to emerging topological data analysis techniques, which leverage persistent homology to capture global shape invariants and connectivity patterns that traditional statistics often overlook [4]. These sections collectively establish the foundational vocabulary necessary for understanding how manual feature extraction encodes biological information.

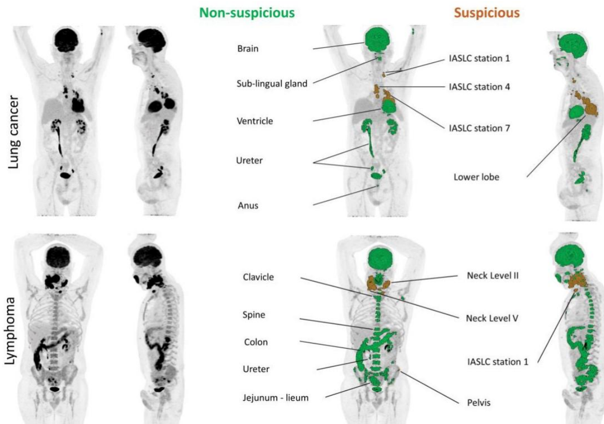

Figure 4: Chart from 'AI-Based Detection, Classification and Prediction/Prognosis in Medical Imaging: Towards Radiophenomics' (arXiv:2110.10332)

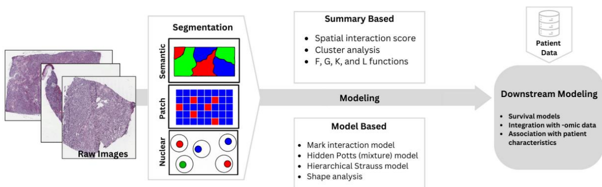

Figure 5: Chart from 'Statistical Analysis of Quantitative Cancer Imaging Data' (arXiv:2409.08809)

Subsequently, the analysis transitions to the classical machine learning frameworks that utilize these engineered features for predictive modeling. This includes a detailed evaluation of support vector machines and kernel-based methods, which excel in high-dimensional spaces with limited sample sizes by maximizing geometric margins [5]. The review also scrutinizes ensemble learning strategies, such as random forests and gradient boosting, highlighting their capacity to mitigate overfitting through bagging and boosting mechanisms [6]. Additionally, the utility of regularized logistic regression is assessed, emphasizing its role in producing sparse, interpretable models that balance predictive performance with clinical transparency, particularly when integrated with clinical covariates.

$$\mathrm{FPR} = \frac{\mathrm{TP}}{\mathrm{TN+FP}}$$

Figure 6: Chart from 'Combining Radiomics and Machine Learning Approaches for Objective ASD Diagnosis: Verifying White Matter Associations with ASD' (arXiv:2405.16248)

To address the inherent challenges of high dimensionality and redundancy in radiomic data, the survey further investigates rigorous feature selection and dimensionality reduction strategies. It details filter methods that operate independently of learning algorithms to prune non-informative variables based on intrinsic statistical properties, with a specific focus on mutual information maximization for capturing non-linear dependencies. The narrative then progresses to compare these classical pipelines with deep learning architectures, analyzing how convolutional neural networks automate feature discovery while discussing hybrid approaches that fuse handcrafted and learned representations. Finally, the review addresses critical issues regarding reproducibility, standardization initiatives, and the translation of these models into real-world clinical settings before outlining future research directions.

The primary contribution of this survey lies in its holistic synthesis of classical radiomics and deep learning, offering a structured comparison that bridges the gap between theoretical algorithm development and clinical implementation [7]. Unlike previous reviews that often treat these domains in isolation, this work explicitly maps the interdependencies between feature engineering paradigms and classification frameworks, providing a clear roadmap for selecting appropriate methodologies based on data availability and clinical objectives. By critically assessing the trade-offs between model interpretability and predictive power, this paper serves as an essential reference for developing next-generation imaging biomarkers that are both statistically robust and clinically viable for improving patient outcomes in oncology.

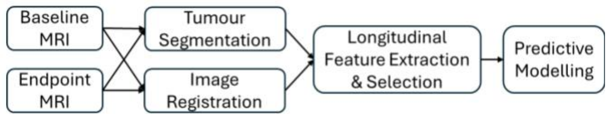

Figure 7: Chart from 'Breast Cancer Neoadjuvant Chemotherapy Treatment Response Prediction Using Aligned Longitudinal MRI and Clinical Data' (arXiv:2512.17759)

2 Handcrafted Radiomics and Classical Machine Learning Pipelines

2.1 Feature Engineering and Extraction Paradigms

2.1.1 Statistical Texture Descriptors and Morphological Metrics

Statistical texture descriptors quantify the spatial distribution of image intensities to characterize tissue heterogeneity beyond simple histogram analysis. Foundational approaches include the Gray-Level Co-occurrence Matrix (GLCM), which evaluates pixel pair correlations to derive metrics such as entropy and contrast, and the Gray-Level Run Length Matrix (GLRLM), which assesses the length of consecutive pixels with identical intensities [8]. These higher-order statistics are frequently complemented by Gray-Level Size Zone Matrices (GLSZM) and Neighboring Gray-Level Dependence Matrices (NGLDM), providing a comprehensive numerical representation of structural patterns like roughness, regularity, and directional dependence within the region of interest.

$$LDE = \sum_{i=1}^{N_g} \sum_{j=1}^{N_d} p(i,j) * j^2$$

Figure 8: Chart from 'MRI Radiomics for IDH Genotype Prediction in Glioblastoma Diagnosis' (arXiv:2409.16329)

Morphological metrics focus on the geometric properties and topological characteristics of segmented lesions, offering critical insights into tumor shape and boundary complexity. These descriptors encompass volumetric measurements, sphericity, compactness, and elongation, alongside advanced surface curvature analyses. By employing Mathematical Morphology operations, researchers can extract shape-based texture features that exploit spatial relationships among pixels, effectively capturing irregular growth patterns and infiltration margins that often correlate with malignant phenotypes and aggressive biological behavior.

To ensure reproducibility and clinical utility, these features are increasingly standardized under initiatives like the Image Biomarker Standardization Initiative (IBSI), which defines rigorous computation protocols across diverse imaging modalities. Feature extraction is often performed on original images as well as filtered variants, including wavelet decompositions and Laplacian of Gaussian transformations, to capture multi-scale information. Subsequent feature selection techniques, such as Minimum Redundancy Maximum Relevance (mRMR), are essential to identify optimal subsets of texture and morphological variables, thereby enhancing the predictive performance of machine learning models for diagnosis, staging, and survival analysis while mitigating the curse of dimensionality.

2.1.2 Wavelet Transformations and Frequency Domain Analysis

Wavelet transformations serve as a critical mechanism in radiomics for decomposing medical images into distinct frequency sub-bands, thereby enabling the extraction of multi-scale textural features that are often imperceptible in the spatial domain. By applying discrete wavelet transforms, original images are filtered through combinations of low-pass and high-pass operators across three dimensions, generating derived representations such as LL, LH, HL, and HH. This decomposition isolates specific structural details, where high-frequency components capture edges and fine textures indicative of tumor heterogeneity, while low-frequency components preserve coarse anatomical structures and overall intensity distributions.

The integration of frequency domain analysis significantly enriches the feature space by allowing the computation of higher-order statistics, including Gray Level Co-occurrence Matrix (GLCM), Gray Level Size Zone Matrix (GLSZM), and Gray Level Dependence Matrix (GLDM) features, on these filtered images. Unlike standard first-order statistics that describe voxel intensity distributions, wavelet-derived features quantify the spatial relationships and patterns of intensity variations at multiple resolutions [9]. Specific filter configurations, such as wavelet-HHH or wavelet-HLH, target unique frequency signatures, facilitating the identification of regions with significant structural detail or noise that correlate strongly with pathological states and treatment responses.

$$\mathrm{CCC} = \frac{\sigma_s^2}{\sigma_s^2 + \sigma_t^2 + \sigma_e^2}.$$

Figure 9: Chart from 'Finding Reproducible and Prognostic Radiomic Features in Variable Slice Thickness Contrast Enhanced CT of Colorectal Liver Metastases' (arXiv:2501.11221)

However, the efficacy of wavelet-based feature extraction is highly dependent on preprocessing parameters, particularly gray-level discretization and bin width selection, which influence feature reproducibility and robustness. While the expansion of the feature set through various wavelet filters enhances the discriminative power of machine learning models, it also introduces computational complexity and the risk of overfitting due to the high dimensionality of the resulting data. Consequently, rigorous feature selection methods, such as LASSO or gradient boosting, are essential to isolate the most informative frequency bands and ensure that the extracted radiomic signatures provide reliable, interpretable insights for clinical decision-making.

2.1.3 Topological Data Analysis and Persistent Homology

Topological Data Analysis (TDA) has emerged as a robust framework for quantifying complex morphological patterns in medical imaging by interpreting images as point clouds rather than conventional pixel arrays [4]. Unlike traditional radiomic features that rely on intensity histograms or texture matrices, TDA leverages algebraic topology to extract tractable invariants that characterize the global shape and connectivity of lesions. This approach is particularly effective in capturing high-order structural heterogeneity, such as vascularization, necrosis, and distinct cell clone distributions, which often evade detection by standard second-order statistics.

Central to this methodology is persistent homology, a technique designed to capture multi-scale topological features across varying resolution thresholds. By applying distance transforms to segmented tumor volumes, persistent homology tracks the birth and death of topological components, such as connected regions, loops, and voids, as the scale parameter changes. This process generates persistence diagrams or barcodes that quantify shape patterns irrespective of absolute size, providing a compact representation of the data's underlying geometric structure that remains invariant to rigid transformations and minor deformations.

Recent applications demonstrate the utility of these topological descriptors in characterizing tumor progression and predicting time-to-event outcomes from MRI scans. By encoding fine and coarse structural details into stable algebraic signatures, persistent homology complements deep learning architectures and classical radiomics, offering enhanced sensitivity to subtle morphological changes associated with disease aggressiveness. The integration of these topological features into predictive modeling pipelines addresses the limitations of voxel-wise analysis, enabling a more comprehensive assessment of intratumoral heterogeneity and improving the reproducibility of prognostic biomarkers across diverse imaging protocols.

2.2 Classical Classification and Regression Frameworks

2.2.1 Support Vector Machines and Kernel Based Methods

Support Vector Machines (SVM) constitute a foundational supervised learning paradigm in radiomics, primarily employed to construct optimal hyperplanes that maximize the margin between distinct diagnostic classes within high-dimensional feature spaces [10]. Originally designed for binary classification, SVMs utilize structural risk minimization to mitigate overfitting, a critical advantage when analyzing datasets where the number of extracted textural and higher-order statistical features significantly exceeds the sample size. The algorithm's efficacy relies heavily on regularization parameters, which balance the trade-off between maximizing the geometric margin and minimizing classification errors, thereby ensuring robust generalization across diverse cancer entities and imaging modalities.

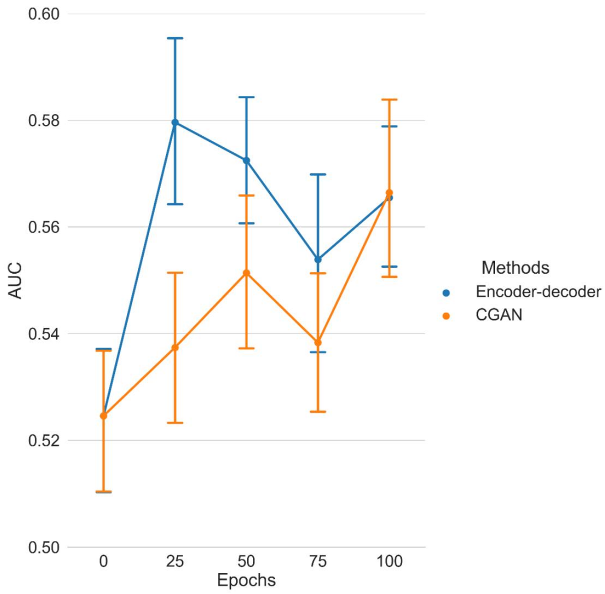

Figure 10: Chart from 'Generative Models Improve Radiomics Performance in Different Tasks and Different Datasets: An Experimental Study' (arXiv:2109.02252)

To address non-linearly separable data distributions inherent in complex medical imagery, kernel-based methods extend the standard linear SVM by implicitly mapping input features into higher-dimensional spaces via the kernel trick. The Radial Basis Function (RBF) kernel is predominantly utilized in survival prediction and tumor classification tasks due to its capacity to model intricate decision boundaries without explicit computation of the transformed coordinates. This approach allows for the effective integration of heterogeneous radiomic signatures, including first-order statistics, shape descriptors, and texture matrices derived from wavelet or Laplacian of Gaussian filters, facilitating the discovery of latent patterns that linear models often fail to capture.

In practical radiomic workflows, SVMs are frequently coupled with recursive feature elimination (RFE) to identify optimal subsets of discriminatory features, enhancing model interpretability and computational efficiency. Comparative studies indicate that while ensemble methods like Random Forests may occasionally yield superior accuracy, SVMs with appropriate kernel selection and hyperparameter tuning via cross-validation remain highly competitive, particularly in scenarios involving small, imbalanced cohorts. Furthermore, the adaptation of SVMs into Support Vector Regression (SVR) frameworks enables continuous outcome prediction, such as estimating patient survival time, solidifying their role as versatile tools in both diagnostic and prognostic clinical decision support systems [11].

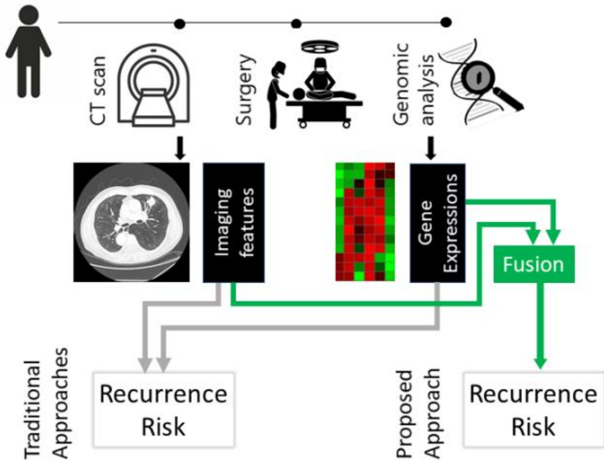

Figure 11: Chart from 'MULTIMODAL FUSION OF IMAGING AND GENOMICS FOR LUNG CANCER RECURRENCE PREDICTION' (arXiv:2002.01982)

2.2.2 Ensemble Learning with Random Forests and Gradient Boosting

Ensemble learning strategies, particularly Random Forests (RF) and Gradient Boosting, have become pivotal in radiomics for mitigating overfitting and enhancing predictive robustness across diverse oncological applications. RF architectures typically aggregate dozens of decision trees trained on bootstrapped subsets of data with random feature initialization, employing soft voting mechanisms to unify outputs for tasks ranging from survival stratification to genetic mutation prediction. This bagging approach effectively reduces variance, as evidenced by studies utilizing fifty-classifier ensembles to predict IDH mutation status and overall survival, where uniform weighting schemes consistently outperformed single-model baselines in handling high-dimensional, redundant radiomic features.

Gradient Boosting frameworks, including XGBoost, LightGBM, and CatBoost, offer a complementary boosting paradigm that sequentially corrects errors from preceding weak learners to minimize bias [12]. These methods excel in managing heterogeneous data types and complex nonlinear relationships inherent in multimodal imaging fused with clinical records. Recent implementations demonstrate that gradient boosting decision trees can achieve superior discrimination capabilities, such as optimizing Area Under the Curve metrics for liver metastasis detection and colorectal polyp classification, often surpassing traditional support vector machines when integrated with wrapper-based feature selection techniques.

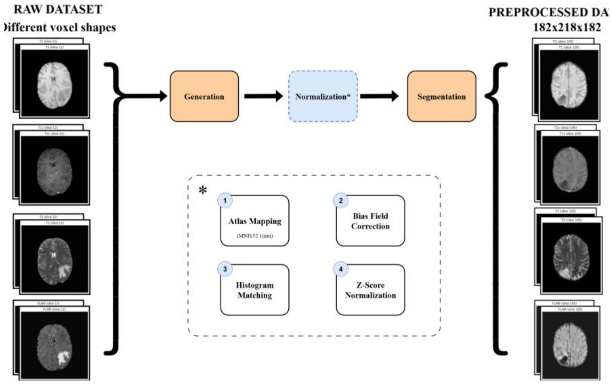

Figure 12: Chart from 'Predicting Brain Tumor Response using a Hybrid Deep Learning and Radiomics Approach' (arXiv:2509.06511)

The efficacy of these ensemble methods is further amplified when combined with rigorous feature engineering pipelines, such as Principal Component Analysis or variable hunting algorithms, to address multicollinearity. Comparative analyses indicate that while deep learning offers automatic feature extraction, ensemble models remain preferred for smaller sample sizes due to their interpretability and stability. By leveraging both bagging and boosting mechanisms, researchers have successfully developed computer-aided diagnosis systems that deliver high sensitivity and specificity, establishing these ensemble techniques as the gold standard for prognostic biomarker discovery and early cancer detection in current medical literature.

2.2.3 Regularized Logistic Regression and Linear Models

Regularized logistic regression serves as a foundational linear classifier in radiomics, particularly effective when managing high-dimensional feature spaces relative to limited sample sizes. By integrating penalty terms into the loss function, these models mitigate overfitting and enhance generalizability across diverse clinical tasks, such as predicting lymph node metastasis or treatment response. The regularization mechanism constrains coefficient magnitudes, ensuring that the model remains robust against noise inherent in medical imaging data while maintaining interpretability of feature contributions.

Among regularization strategies, the Least Absolute Shrinkage and Selection Operator (LASSO) employs an L_1 penalty to induce sparsity, effectively performing automatic feature selection by shrinking irrelevant coefficients to zero [8]. This parsimonious approach is critical for identifying stable biomarkers from hundreds of radiomic features, often yielding compact models that rival complex deep learning architectures in specific diagnostic scenarios. Conversely, Ridge regression utilizes an L_2 penalty to distribute weight across correlated features without eliminating them, which is advantageous when multicollinearity exists among texture or morphological descriptors. Elastic Net combines both penalties to balance feature selection with group effect preservation, offering a flexible framework for optimizing predictive performance in survival analysis and tumor progression modeling.

Comparative studies frequently demonstrate that regularized linear models achieve competitive Area Under the Curve (AUC) scores against nonlinear alternatives like Support Vector Machines or gradient boosting trees, especially when feature engineering is rigorous. While deep learning methods excel in raw image processing, regularized logistic regression remains preferred for tabular radiomic data due to its computational efficiency and transparent decision boundaries. The integration of these linear models with clinical factors often results in hybrid nomograms that provide superior calibration and discriminative ability, validating their enduring relevance in developing clinically deployable AI tools for oncology.

2.3 Feature Selection and Dimensionality Reduction

2.3.1 Filter Methods and Mutual Information Maximization

Filter methods constitute a fundamental stage in the radiomics pipeline, operating independently of specific learning algorithms to evaluate feature relevance through intrinsic statistical properties [13]. These techniques efficiently prune high-dimensional data spaces by ranking features based on metrics such as variance, correlation, or mutual information before model training begins. By eliminating non-informative variables, such as those with zero median absolute deviation, filter methods significantly reduce computational complexity and mitigate the risk of overfitting, which is particularly critical when dealing with heterogeneous medical imaging data where noise can obscure biological signals.

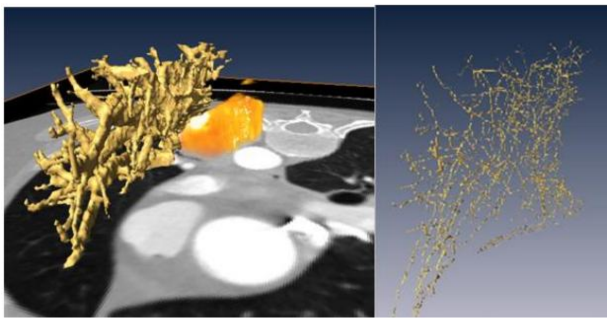

Figure 13: Chart from 'A DEEP LEARNING-FACILITATED RADIOMICS SOLUTION FOR THE PREDICTION OF LUNG LESION SHRINKAGE IN NON-SMALL CELL LUNG CANCER TRIALS' (arXiv:2003.02943)

Mutual Information (MI) maximization serves as a robust theoretical framework within this category, quantifying the dependency between extracted features and target response variables without assuming linear relationships. Unlike simple correlation coefficients, MI captures higher-order statistical dependencies, making it exceptionally suitable for analyzing complex texture patterns derived from wavelet or Laplacian of Gaussian filtered images. The core objective is to identify a feature subset that maximizes the mutual information with the clinical outcome, thereby ensuring that the selected descriptors retain the maximum amount of predictive information regarding tumor grade, survival rates, or genetic mutations.

To address the inherent redundancy in radiomic features, the Minimum Redundancy Maximum Relevance (mRMR) criterion extends basic MI maximization by optimizing a dual objective function. This approach selects features that exhibit strong relevance to the target class while simultaneously minimizing the pairwise mutual information among the selected features themselves. By balancing these competing goals, mRMR constructs a compact and discriminatory feature set that avoids multicollinearity, ultimately enhancing the generalizability and interpretability of downstream machine learning classifiers used in precision oncology applications.

2.3.2 Wrapper Strategies and Recursive Feature Elimination

Wrapper strategies address the limitations of filter methods by evaluating feature subsets through the specific performance of a predictive model, thereby directly optimizing for classification accuracy or regression error. Unlike embedded techniques that perform selection during model training, wrapper approaches treat feature selection as a search problem, iteratively adding or removing variables to identify the optimal combination. Although computationally intensive due to the repeated model retraining required for each subset evaluation, these methods effectively capture complex feature interactions and are particularly valuable in radiomics where feature redundancy and high dimensionality often lead to overfitting.

Recursive Feature Elimination (RFE) stands as the most prominent wrapper algorithm, employing a greedy backward elimination strategy to rank feature importance [14]. The process initiates by training a classifier, typically a Support Vector Machine (SVM), on the complete feature set to compute weights or importance scores. The feature with the lowest contribution to the decision boundary is systematically pruned, and the model is retrained on the reduced subset. This cycle repeats recursively until a predefined number of features remains or performance metrics plateau, ensuring that the final subset consists of variables that collectively maximize predictive power while minimizing model complexity.

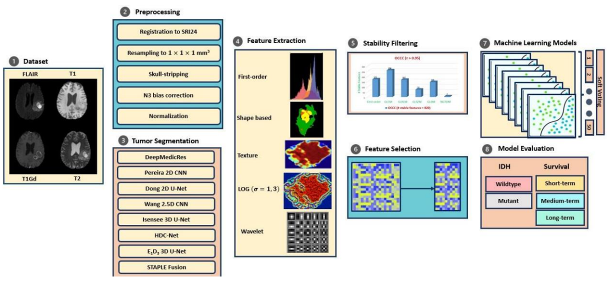

Figure 14: Chart from 'Towards robust radiomics and radiogenomics predictive models for brain tumor characterization' (arXiv:2406.03583)

Empirical applications in medical imaging demonstrate RFE's efficacy in distilling large radiomic signatures into compact, interpretable panels. Studies utilizing RFE-SVM have successfully identified optimal subsets comprising first-order and texture features while discarding unstable shape descriptors, often yielding robust Area Under the Curve (AUC) values despite significant dimensionality reduction. By integrating cross-validation within the elimination loop, researchers mitigate the risk of Type I errors and ensure that selected features maintain stability across different data splits. Consequently, RFE serves as a critical mechanism for balancing model generalizability with the clinical need for parsimonious, explainable predictive tools.

2.3.3 Embedded Techniques via LASSO and Elastic Net

Embedded feature selection techniques, particularly Least Absolute Shrinkage and Selection Operator (LASSO) and Elastic Net, have become indispensable in radiomics for managing high-dimensional data while mitigating overfitting [15]. LASSO employs an L1-norm regularization penalty within the regression objective function, effectively shrinking the coefficients of irrelevant features to exactly zero [16]. This inherent sparsity induction performs simultaneous variable selection and parameter estimation, isolating a compact subset of robust descriptors from thousands of extracted textural and higher-order statistical features without requiring a separate filtering stage.

$$L = L_{seg} + L_{surv}$$

Figure 15: Chart from 'Radiomics-enhanced Deep Multi-task Learning for Outcome Prediction in Head and Neck Cancer' (arXiv:2211.05409)

$$\min_{\omega} \sum_{n=1}^{N} (y_n - \omega^T x_n)^2 + \lambda \|\omega\|_1$$

Figure 16: Chart from '2D and 3D CT Radiomic Features Performance Comparison in Characterization of Gastric Cancer: A Multi-center Study' (arXiv:2210.16640)

While LASSO excels at producing sparse models, it exhibits limitations when dealing with highly correlated features, often arbitrarily selecting one variable from a correlated group while discarding others. Elastic Net addresses this constraint by combining L1 and L2 regularization penalties, thereby inheriting the sparsity of

LASSO and the grouping effect of Ridge regression. This hybrid approach ensures that strongly correlated radiomic signatures, such as those derived from Gray Level Co-occurrence Matrix or wavelet transformations, are retained or eliminated together, leading to more stable and interpretable predictive models in multi-modal clinical settings.

The implementation of these embedded methods typically involves rigorous cross-validation protocols to tune regularization hyper-parameters, ensuring that feature selection occurs strictly within training folds to prevent information leakage. By integrating these techniques directly into the model training process, researchers can derive parsimonious radiomic signatures that maintain high predictive accuracy across diverse cohorts. Consequently, LASSO and Elastic Net serve as critical preprocessing mechanisms that enhance the generalizability of downstream machine learning classifiers, such as Support Vector Machines and linear regression models, in diagnostic and prognostic applications.

2.4 Data Harmonization and Robustness Assessment

2.4.1 Batch Effect Correction and ComBat Harmonization

Multicenter radiomics studies frequently suffer from batch effects arising from heterogeneous imaging protocols, scanner vendors, and acquisition parameters, which introduce non-biological variance that obscures genuine phenotypic signals [17]. These technical artifacts severely compromise the reproducibility and generalizability of predictive models, necessitating robust harmonization strategies prior to feature analysis. Without effective correction, models trained on data from specific institutions often fail to generalize to external cohorts, limiting clinical translation and the reliability of quantitative biomarkers across diverse healthcare settings.

ComBat harmonization, adapted from genomics, has emerged as a predominant statistical method for mitigating these batch effects in the feature domain. This empirical Bayes framework adjusts feature distributions by estimating and removing additive and multiplicative scanner-specific effects while preserving biological variability associated with clinical outcomes. By modeling batch effects as location and scale

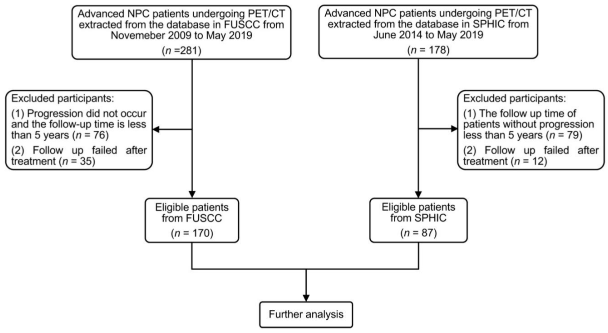

Figure 17: Chart from 'Prediction of 5-year Progression-Free Survival in Advanced Nasopharyngeal Carcinoma with Pretreatment PET/CT using MultiModality Deep Learning-based Radiomics' (arXiv:2103.05220)

shifts, ComBat aligns feature distributions across different sites without requiring raw image reprocessing, making it computationally efficient for large-scale retrospective studies involving hundreds of radiomic features extracted under varying conditions.

Recent advancements have extended ComBat to handle non-linear batch effects and integrate covariates directly into the harmonization model, ensuring that critical biological factors such as tumor stage or molecular status are not inadvertently removed during correction. Comparative analyses indicate that ComBat often outperforms traditional normalization techniques like Z-score standardization in restoring feature stability, as evidenced by improved Intraclass Correlation Coefficients between test-retest scans [18]. However, careful validation remains essential, as over-correction can potentially eliminate subtle but clinically relevant imaging patterns, necessitating a balanced approach that maximizes technical consistency while retaining diagnostic fidelity.

2.4.2 Segmentation Variability and Intraclass Correlation

Segmentation variability in colorectal cancer imaging arises from significant morphological heterogeneity, per-patient anatomical diversity, and artifacts induced by bowel movements. These factors introduce non-physiological noise into predictive models, particularly when relying on fully automatic convolutional neural networks that may produce inconsistent boundary delineations. The resulting instability in segmentation maps directly compromises the reliability of downstream radiomic feature extraction, as minor deviations in tumor contours can drastically alter calculated texture and shape descriptors [19].

To mitigate these discrepancies, stability filtering techniques have been employed to minimize variability, demonstrated by an order-of-magnitude reduction in the Relative Standard Deviation of model performance metrics. Without such interventions, the intraclass correlation among features extracted from repeated or slightly perturbed segmentations remains low, undermining the reproducibility of biomarkers. This sensitivity is especially pronounced in high-dimensional feature spaces where first-order statistics and higher-order texture matrices, such as the Gray-Level Co-occurrence Matrix, exhibit strong dependence on precise voxel inclusion within the region of interest.

Furthermore, the lack of absolute agreement between manual readers necessitates robust automated methods that can generalize across varying expertise levels and imaging conditions. While deep learning architectures like U-Net variants offer promising accuracy, their output variability must be rigorously quantified to ensure clinical utility. Addressing segmentation uncertainty through ensemble approaches or probabilistic modeling is critical for establishing high intraclass correlation coefficients, thereby ensuring that extracted features reflect true biological variation rather than algorithmic inconsistency or annotation noise.

2.4.3 Resampling Strategies and Acquisition Parameter Stability

Resampling serves as a critical preprocessing step to standardize voxel spacing across heterogeneous imaging protocols, ensuring that texture features reflect consistent physical distances rather than variable pixel dimensions. While guidelines often recommend isotropic resampling to facilitate comparability, empirical evidence suggests this process has limited efficacy in fully mitigating the effects of varying slice thicknesses or inherent resolution disparities. Consequently, reliance on standard interpolation methods alone may introduce artifacts or alter feature distributions, particularly when original acquisition parameters differ significantly across scanner manufacturers and field strengths.

To address acquisition parameter instability, harmonization techniques such as ComBat are frequently employed to adjust feature distributions for variations arising from different hospitals, scanner types, or protocol settings. However, even after statistical harmonization, performance degradation often persists in external validation cohorts due to domain shifts caused by distinct radiotracers, sequence parameters, or hardware configurations. This indicates that while resampling and harmonization reduce inter-scanner variability, they do not entirely eliminate the dependency of radiomic signatures on specific acquisition environments, necessitating robust validation across diverse datasets.

Advanced strategies increasingly combine resampling with adaptive deep learning architectures to enhance robustness under degraded conditions, such as reduced input resolution or additive Gaussian noise. Experiments demonstrate that model accuracy declines sharply as spatial resolution decreases, highlighting the sensitivity of convolutional networks to resampling-induced information loss. Therefore, future frameworks must integrate domain-adaptive learning and optimized resampling protocols that preserve geometric fidelity while minimizing the introduction of interpolation bias, ensuring reliable generalization across low-resource and multi-center imaging scenarios.

3 Deep Learning Architectures for Multimodal Fusion and Representation Learning

3.1 Advanced Neural Network Topologies

3.1.1 Convolutional Neural Networks and Residual Architectures

Convolutional Neural Networks (CNNs) have become the cornerstone of medical image analysis, automating the extraction of hierarchical features from raw data to replace traditional handcrafted descriptors [20]. Standard architectures utilize sequential convolution, pooling, and fully connected layers to identify patterns ranging from low-level edges to complex anatomical structures in modalities such as CT, MRI, and histopathology. However, as network depth increases to capture more abstract representations, standard CNNs often encounter the vanishing gradient problem and performance degradation, limiting their efficacy in deep configuration scenarios.

To address these limitations, Residual Networks (ResNets) introduced skip connections that facilitate identity mapping, allowing gradients to bypass intermediate layers and enabling the training of networks exceeding one hundred layers without accuracy loss. This residual learning mechanism effectively mitigates gradient vanishing and model degradation, making deep architectures viable for complex diagnostic tasks like tumor detection and lung nodule screening [21]. Further advancements, such as ResNeXt, incorporate aggregated residual transformations to enhance representational capacity through cardinality, while maintaining computational efficiency and stability during backpropagation.

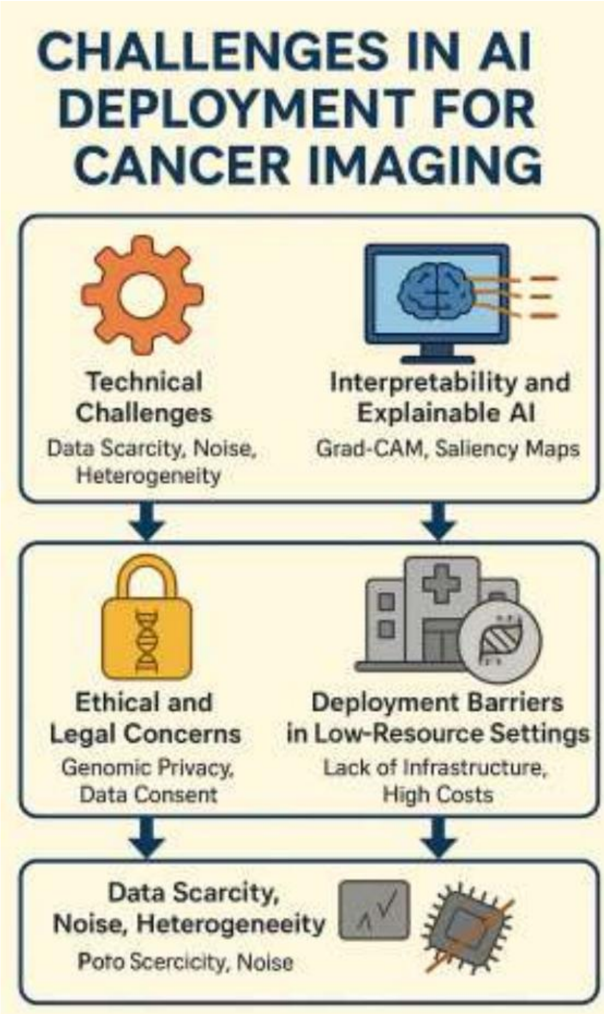

Figure 18: Chart from 'Deep Learning-Based Computer Vision Models for Early Cancer Detection Using Multimodal Medical Imaging and Radiogenomic Integration Frameworks' (arXiv:2512.00714)

In practice, variants like ResNet-50 and 3D ResNet are extensively employed for end-to-end learning in predictive modeling, often serving as backbone encoders for segmentation frameworks like U-Net. These architectures typically integrate batch normalization and non-linear activation functions, such as ReLU or PReLU, within residual blocks to accelerate convergence and improve generalization. By leveraging transfer learning from large-scale natural image datasets, these residual architectures significantly reduce the data hunger of deep models, demonstrating superior performance in classifying lesions and predicting patient survival compared to shallower counterparts.

3.1.2 Vision Transformers and Self Attention Mechanisms

Vision Transformers (ViT) represent a paradigm shift in medical image analysis by replacing convolutional inductive biases with self-attention mechanisms that process images as sequences of patches. Unlike traditional Convolutional Neural Networks (CNNs) constrained by fixed local receptive fields, ViTs dynamically capture long-range dependencies and global contextual relationships across the entire image. This capability is particularly advantageous for analyzing heterogeneous pathologies, such as complex tumors, where spatial correlations between distant regions are critical for accurate characterization and diagnosis.

The core of this architecture lies in the Multi-Head Self-Attention (MSA) mechanism, which computes attention weights by mapping input features into query, key, and value matrices [22]. This process allows the model to jointly attend to information from different representation subspaces, effectively weighing the importance of specific image regions relative to others while suppressing irrelevant background noise. Advanced variants, such as Swin Transformers, further optimize this approach by incorporating hierarchical windows and shifted attention strategies, thereby balancing computational efficiency with the ability to model multi-scale features essential for high-resolution biomedical imaging.

In multimodal contexts like radiogenomics, transformer-based encoders extend these principles by fusing imaging patches with genomic token embeddings through unified attention layers. This facilitates the simultaneous learning of cross-modal interactions, enabling the model to identify subtle correlations between phenotypic visual patterns and underlying genotypic markers. By leveraging unsupervised pre-training and end-to-end optimization, these attention-driven architectures significantly enhance feature representation, offer-

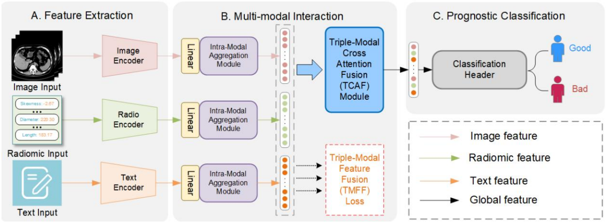

Figure 19: Chart from 'TMI-CLNET: TRIPLE-MODAL INTERACTION NETWORK FOR CHRONIC LIVER DISEASE PROGNOSIS FROM IMAGING, CLINICAL, AND RADIOMIC DATA FUSION' (arXiv:2502.00695)

ing superior performance in tasks ranging from lesion segmentation to predictive outcome modeling compared to conventional deep learning baselines.

3.1.3 Graph Neural Networks for Structural Representation

Graph Neural Networks (GNNs) have emerged as a pivotal architecture for modeling complex structural relationships in radiogenomics, effectively addressing the limitations of standard convolutional networks in capturing non-Euclidean data dependencies. By representing genes, radiomic descriptors, and clinical attributes as nodes within a graph, GNNs encode biological and imaging interdependencies through edge connections, facilitating automatic feature discovery without manual engineering. This structural representation is particularly advantageous in scenarios characterized by sparse labeled data, where traditional 3D approaches often suffer from the curse of dimensionality and overfitting.

Advanced implementations leverage population graph strategies to construct latent graphs based on pairwise patient similarities derived from multimodal inputs such as MRI sequences and genomic profiles. Architectures like Deep Patient Graph Convolutional Networks utilize these structures to identify distinct patient subgroups and predict survival outcomes with superior accuracy compared to non-graph-based methods. To enhance model expressiveness and mitigate gradient vanishing in deep configurations, recent designs incorporate residual connections, node feature normalization, and attention mechanisms that dynamically weigh the importance of neighboring nodes during message passing.

Furthermore, GNNs enable sophisticated fusion strategies that correlate radiomic phenotypes with genomic signatures, supporting early diagnosis across various malignancies including glioblastoma and breast cancer. The integration of hypergraph neural networks extends this capability by modeling higher-order interactions beyond simple pairwise connections, thereby capturing more intricate biological pathways. Despite challenges in defining optimal graph topologies from heterogeneous medical data, these models provide a robust framework for interpretable deep learning, offering critical insights into biomarker identification and therapeutic response prediction through structured data representation.

3.2 Multimodal Integration Strategies

3.2.1 Early Fusion and Concatenation Based Approaches

Early fusion strategies integrate raw imaging data or extracted feature vectors from distinct modalities, such as PET and CT, at the input stage prior to model processing. By concatenating these multi-source inputs, the approach aims to preserve intrinsic spatial and textural correlations that exist between complementary imaging domains. This direct combination allows deep learning architectures to learn joint representations immediately, theoretically capturing deep interdependencies that might be lost in later processing stages.

The primary mechanism within this paradigm involves channel-wise or vector-level concatenation, where multimodal data is merged into a unified tensor. While this method simplifies the network architecture by requiring a single stream for feature extraction, it imposes strict requirements on data alignment and resolution consistency across modalities. Misregistration or varying voxel sizes can introduce noise that degrades model performance, as the network may struggle to distinguish biological signals from artifacts caused by spatial discrepancies in the concatenated input.

Empirical evaluations indicate that while concatenation can enhance prognostic accuracy by providing comprehensive tumor characterization, its efficacy is highly dataset-dependent. In certain contexts, simply appending modalities yields marginal improvements over single-modality baselines or even leads to convergence instability if the feature scales are not properly normalized. Consequently, successful implementation often necessitates rigorous preprocessing, including affine transformations and intensity standardization, to ensure that the fused input effectively informs the downstream classification or survival analysis tasks without overwhelming the model with redundant information.

3.2.2 Late Fusion and Decision Level Aggregation

Late fusion, or decision-level aggregation, operates by independently processing distinct data modalities through separate models before combining their outputs at the final prediction stage. This architectural strategy offers significant flexibility, allowing for the integration of heterogeneous data sources such as imaging, genomics, and electronic health records without requiring strict spatial alignment or uniform resolution across inputs. By merging classification scores or risk probabilities via ensemble techniques like weighted averaging or stacking, late fusion effectively mitigates the sensitivity to misalignment often observed in early fusion approaches, thereby enhancing model robustness in multi-center clinical studies.

Despite its adaptability, late fusion inherently limits the capture of deep inter-modal dependencies, as it ignores low-level feature correlations that exist prior to the decision boundary. Consequently, while this method excels in scenarios where modalities possess disparate statistical distributions or acquisition protocols, it may underperform in tasks requiring intricate spatial-textural synergy, such as precise tumor segmentation. To address these limitations, recent advancements have introduced hybrid frameworks that blend late fusion with intermediate layer integration, attempting to balance the preservation of modality-specific nuances with the benefits of high-level feature correlation.

In the context of oncological prognosis, late fusion has demonstrated efficacy in risk stratification and survival analysis by aggregating predictions from specialized sub-networks. Applications range from predicting head and neck cancer recurrence using PET-CT radiomics to assessing lung cancer outcomes through co-learning feature maps. Although standalone late fusion models may occasionally yield lower concordance indices compared to deeply integrated architectures, their interpretability and ability to incorporate diverse clinical variables make them indispensable for developing transparent, clinically actionable decision support systems that facilitate improved patient staging and treatment planning.

3.2.3 Cross Modal Attention and Tensor Fusion

Cross-modal attention mechanisms have emerged as a pivotal strategy for dynamically integrating heterogeneous medical imaging data, addressing the limitations of static concatenation. Unlike early or late fusion approaches that may overlook deep interdependencies, attention-gated networks learn to recalibrate feature maps by assigning context-dependent weights to specific modalities. This allows the architecture to suppress irrelevant background noise while amplifying salient tumor-specific features across different imaging domains, such as PET and CT, thereby enhancing localization accuracy and robustness against modality-specific artifacts.

Beyond simple attention weighting, tensor fusion techniques offer a more rigorous mathematical framework for modeling high-order interactions between multimodal representations. Methods such as multimodal compact bilinear pooling project features from separate streams into a joint embedding space, capturing complex non-linear correlations that linear fusion layers often miss. By utilizing outer products or low-rank approximations, these models generate comprehensive joint representations that preserve the intricate relationships between structural and functional data, which is critical for tasks requiring fine-grained prognostication and survival analysis.

The synergy between attention mechanisms and tensor fusion has demonstrated superior performance in benchmark segmentation and classification tasks compared to single-modality baselines. Advanced architectures now incorporate skip connections with attention blocks to refine decoder outputs, ensuring that fused features maintain spatial fidelity during upsampling. Furthermore, the integration of these fusion strategies with multi-task learning objectives enables simultaneous optimization of segmentation, classification, and survival prediction, leveraging the complementary nature of multimodal data to achieve state-of-the-art diagnostic accuracy and generalizability across diverse clinical cohorts.

3.3 Weakly Supervised and Multiple Instance Learning

3.3.1 Attention Based MIL Pooling Mechanisms

Attention-based Multiple Instance Learning (MIL) pooling mechanisms address the limitations of traditional aggregation functions, such as max or mean pooling, by introducing trainable parameters to dynamically weigh instance contributions within a bag [23]. Unlike static operators that assume uniform importance or rely solely on the most extreme feature values, attention-based pooling learns a data-driven distribution over instances. This is typically achieved by projecting instance embeddings through fully connected layers to generate scalar attention scores, which are subsequently normalized via a softmax function. This process allows the model to selectively focus on discriminative regions, such as specific tumor subregions, while suppressing irrelevant background noise, thereby creating a more representative bag-level embedding for downstream prediction tasks.

$$Y = \sum_{n} y_n$$

Figure 20: Chart from 'Metastatic Cancer Outcome Prediction with Injective Multiple Instance Pooling' (arXiv:2203.04964)

Advanced variations of this mechanism extend beyond simple scalar weighting to incorporate multi-head self-attention and hierarchical structures inspired by Transformer architectures. By employing multi-head attention, models can jointly attend to information from different representation subspaces, capturing complex long-range dependencies and heterogeneous patterns often present in medical imaging and genomics data. Furthermore, injective pooling functions, including unnormalized attention and sum pooling, have been proposed to better preserve the cardinality and distributional properties of instances within a bag [23]. These approaches are particularly critical in multifocal disease scenarios where the collective interaction of multiple lesions determines clinical outcomes, a nuance often lost in non-injective traditional pooling methods.

Despite their theoretical advantages, the efficacy of attention-based MIL pooling is highly dependent on dataset scale and architectural design [24]. Empirical studies indicate that while attention mechanisms consistently outperform max pooling in representational power, they may exhibit instability or fail to surpass mean pooling in data-scarce regimes due to the increased number of learnable parameters. The integration of attention modules into deep backbones, such as 3D U-Nets or graph convolutional networks, further enhances feature recalibration across spatial and channel dimensions. However, careful regularization and optimization strategies are required to prevent overfitting, ensuring that the learned attention maps genuinely reflect pathological relevance rather than artifacts of the training distribution.

$$recall = \frac{TP}{TP+TN}$$

Figure 21: Chart from 'Lung Cancer Diagnosis Using Deep Attention Based Multiple Instance Learning and Radiomics' (arXiv:2104.14655)

3.3.2 Patch Level Feature Aggregation and Context Modeling

Patch-level feature aggregation serves as a critical mechanism for transforming local image descriptors into comprehensive global representations, particularly within Multiple Instance Learning (MIL) frameworks where whole scans are treated as bags of patch instances. Traditional pooling strategies, including max, mean, and attention-based mechanisms, are employed to synthesize instance-level embeddings into bag-level predictions, effectively mitigating the noise inherent in redundant or irrelevant patch information. Advanced architectures further refine this process by utilizing hierarchical stages, such as those found in Swin Transformers, where non-overlapping 3D patches are projected into latent spaces and processed through successive blocks to capture multi-scale spatial dependencies without excessive computational overhead.

Context modeling at the patch level is increasingly augmented by deep learning structures designed to preserve fine-grained details while integrating broader semantic information. Encoder-decoder networks facilitate precise localization by concatenating high-resolution features from encoder blocks with upsampled contextual data from decoder paths, often enhanced by instance normalization and spatial dropout to prevent overfitting. Furthermore, the incorporation of squeeze-and-excitation blocks allows for dynamic channel-wise and temporal feature recalibration, ensuring that the model prioritizes informative feature maps across different dimensions. This approach enables the network to learn effective representations from information extending beyond strict ground-truth boundaries, thereby improving robustness against variations in tumor morphology and imaging artifacts.

To address the challenge of feature redundancy and high dimensionality resulting from extensive patch extraction, rigorous dimensionality reduction and selection protocols are essential. Techniques such as Principal Component Analysis, LASSO regularization, and correlation-based filtering are routinely applied to eliminate highly correlated deep and handcrafted features before final model integration. Hybrid fusion strategies often combine these optimized patch-level features with clinical data or genomics profiles, leveraging both low-level textural details and high-level semantic correlations. By aggregating features through global pooling layers or graph convolutional networks that establish relationships between nearest neighbors, these models achieve superior accuracy in tasks ranging from tumor segmentation to survival analysis, balancing the trade-off between local detail preservation and global context understanding.

3.3.3 Slide Group Learning and Holistic Interaction Modeling

Slide group learning addresses the computational constraints of whole slide images by treating them as bags of instances within a Multiple Instance Learning framework [25]. Rather than processing gigapixel images directly, this approach partitions slides into manageable patches, extracting local features that are subsequently aggregated to form a holistic slide-level representation. Advanced pooling mechanisms and attention-based aggregation have replaced traditional max or mean pooling, enabling models to dynamically weigh discriminative regions while suppressing background noise, thereby preserving critical pathological context without exhaustive pixel-wise annotation.

$$P(y) = \frac{1}{1 + \exp(-z)},$$

Figure 22: Chart from ’MIL vs. Aggregation: Evaluating Patient-Level Survival Prediction Strategies Using Graph-Based Learning’ (arXiv:2503.23042)

Holistic interaction modeling extends this paradigm by capturing long-range dependencies and structural relationships between distinct tissue regions that standard convolutional architectures often miss. By leveraging graph neural networks or transformer-based self-attention mechanisms, these methods construct global representations where individual patches interact as nodes within a complex topological structure. This facilitates the learning of spatial configurations and morphological patterns across the entire tissue sample, allowing the model to infer diagnostic signals from the collective arrangement of cellular structures rather than isolated local features.

The integration of slide group learning with holistic interaction modeling significantly enhances prognostic accuracy and generalizability in computational pathology. By jointly optimizing local feature extraction and global context aggregation, these frameworks effectively mitigate the semantic gap between patch-level predictions and slide-level diagnoses. Furthermore, this unified approach supports robust performance across diverse imaging protocols and patient demographics, offering a scalable solution for identifying subtle disease subtypes and predicting clinical outcomes from high-resolution histopathological data.

3.4 Self Supervised and Generative Representation Learning

3.4.1 Contrastive Learning and Momentum Encoders

Contrastive learning has emerged as a pivotal self-supervised paradigm in medical image analysis, addressing the scarcity of annotated data by learning robust representations through instance discrimination. This approach optimizes an encoder to maximize agreement between differently augmented views of the same image while minimizing similarity with other samples in the batch.

By leveraging large-scale unlabeled datasets, these methods effectively capture invariant semantic features essential for downstream tasks such as tumor segmentation and survival prediction, overcoming the limitations of supervised pre-training on natural images which often fail to generalize to specific medical domains.

To stabilize training and expand the effective batch size without prohibitive memory costs, momentum encoders are frequently integrated into contrastive frameworks. In this architecture, a target network is updated as an exponential moving average of the query encoder's parameters rather than through direct backpropagation. This momentum mechanism ensures that the key representations remain consistent over time, providing reliable negative samples for the contrastive loss function. The decoupling of gradient updates allows the model to maintain a large and diverse dictionary of features, which is critical for distinguishing subtle pathological patterns in heterogeneous imaging modalities like MRI and histopathology.

The synergy between contrastive objectives and momentum encoders facilitates the learning of rich, hierarchical feature spaces that are resilient to noise and domain shifts. Recent adaptations have extended this framework to multi-instance learning scenarios, enabling the aggregation of patch-level features into comprehensive slide-level representations for whole-slide image classification [25]. By enforcing consistency across spatial and temporal augmentations, these models achieve superior performance in few-shot settings and demonstrate enhanced capability in disentangling complex biological variations, thereby establishing a new standard for representation learning in computational oncology and diagnostic imaging.

3.4.2 Variational Autoencoders and Latent Space Modeling

Variational Autoencoders (VAEs) have emerged as pivotal architectures for unsupervised representation learning in high-dimensional medical data, effectively addressing the scarcity of labeled datasets. By mapping input modalities, such as multi-channel PET/CT or genomic sequences, into a probabilistic latent space, VAEs enforce continuity and smoothness through Kullback-Leibler divergence regularization. This approach facilitates robust dimensionality reduction while preserving critical biological variance, enabling the synthesis of realistic imaging data for augmentation and the detection of subtle anomalies that deterministic autoencoders might overlook due to overfitting in abridged models.

Beyond simple compression, advanced VAE frameworks integrate domain-specific prior knowledge to enhance interpretability and biological relevance within the latent manifold. Techniques such as Laplacian regularization leverage graphical structures, including protein-protein interaction networks, to constrain the latent geometry, ensuring that related biological entities maintain proximal representations. Furthermore, architectural modifications like sparse connectivity or positive weight constraints in the decoder nullify non-biological associations, allowing the model to isolate latent factors directly correlated with patient prognosis rather than mere reconstruction fidelity.

In predictive modeling workflows, the bottleneck layer of the VAE often serves as a sophisticated feature extractor for downstream survival analysis or classification tasks. By projecting heterogeneous data sources into a unified latent space, these models facilitate effective multimodal fusion, aggregating microenvironmental cues and molecular signals into dense vectors suitable for Cox proportional hazards modeling or clustering. Although simultaneous training of generative and discriminative branches can introduce optimization instability requiring careful hyperparameter tuning, the resulting latent representations significantly improve risk stratification accuracy compared to traditional handcrafted radiomic features.

3.4.3 Generative Adversarial Networks for Data Synthesis

Generative Adversarial Networks (GANs) have emerged as a pivotal solution for addressing data scarcity and class imbalance in medical imaging, particularly for synthesizing positron emission tomography (PET) images from computed tomography (CT) scans. By leveraging multi-channel architectures, these models generate synthetic PET volumes that preserve high-uptake regions critical for lesion detection while constraining anatomical appearance through adversarial training. This cross-modality synthesis effectively augments training datasets, enabling robust model performance even when malignant cases are significantly underrepresented compared to normal samples.

The application of GANs extends beyond simple oversampling techniques like SMOTE, offering a sophisticated mechanism to reduce false positives and enhance the generalizability of deep learning classifiers. In lung cancer and head-and-neck tumor studies, synthetic data generated via GANs has demonstrated the ability to mitigate overfitting in convolutional neural networks, often yielding accuracy rates comparable to those trained on extensive real-world cohorts. Furthermore, these generative models facilitate the creation of artificial datasets for transfer learning, allowing pre-trained networks to adapt to specific diagnostic tasks where annotated data is prohibitively expensive or difficult to acquire.

Despite these advancements, significant challenges remain regarding the dependency on paired datasets for effective training, which necessitates precise pixel-to-pixel correspondence between source and target modalities. While GAN-based methods generally outperform conventional synthesis techniques in producing realistic images with superior quantitative metrics, the computational cost of training these models can be substantial, often requiring days of processing on high-performance hardware. Future research must address the stability of simultaneous training processes and explore unpaired synthesis strategies to broaden the applicability of generative models in multimodal cancer detection systems.

4 Radiomics-Enhanced Deep Learning for Survival Analysis and Prognosis

4.1 Hybrid Feature Fusion Architectures

4.1.1 Disentangled Representation Learning for Complementary Features

Disentangled representation learning has emerged as a critical paradigm for isolating complementary features within multimodal medical data, addressing the limitations of naive concatenation strategies that often fail to exploit inter-modal synergies. Traditional deep learning approaches frequently merge imaging and clinical vectors through simple fusion layers, resulting in entangled representations where modality-specific nuances are obscured by dominant signals. By enforcing statistical independence among latent variables, disentangled frameworks decompose high-dimensional inputs into distinct factors of variation, ensuring that unique prognostic indicators from each modality, such as texture patterns in MRI or metabolic activity in PET, are preserved rather than diluted during the integration process.

To achieve this separation, recent methodologies employ variational inference and adversarial training to align shared semantic concepts while pushing modality-unique characteristics into orthogonal subspaces. This architectural design allows the model to explicitly capture complementary information, such as the correlation between tumor morphology and genetic expression, without allowing one data source to overpower the other. Techniques utilizing maximum likelihood estimation and information bottleneck principles further refine these representations by minimizing mutual information between disentangled codes, thereby enhancing the interpretability of the learned features and ensuring that the model focuses on clinically relevant biomarkers rather than spurious correlations inherent in noisy medical datasets.

The application of these disentangled representations significantly improves survival prediction accuracy by enabling robust feature fusion that leverages the full spectrum of available diagnostic information. Unlike standard multi-task learning which implicitly guides feature extraction, explicit disentanglement facilitates the identification of weak but significant signals that conventional radiomics or end-to-end convolutional networks might overlook. Consequently, this approach not only boosts predictive performance in scenarios with censored data but also provides a transparent mechanism for validating how distinct biological processes contribute to patient outcomes, marking a substantial advancement over black-box fusion models in precision oncology.

4.1.2 Joint Optimization of Handcrafted and Deep Descriptors

The integration of handcrafted radiomic features with deep learning descriptors addresses the limitations inherent in relying solely on either approach. While handcrafted features, such as Haralick texture metrics and Fourier shape descriptors, offer interpretable, low-dimensional representations of tumor heterogeneity, they often fail to capture high-level semantic patterns. Conversely, deep convolutional neural networks automatically learn robust feature hierarchies but may struggle with small datasets or lack explicit focus on specific regions of interest without segmentation masks. Joint optimization strategies aim to leverage the complementary strengths of both, enhancing prognostic accuracy by combining engineered stability with data-driven adaptability.

Architecturally, this synergy is frequently realized through multi-task learning frameworks or hybrid fusion layers where distinct sub-models process handcrafted and deep features simultaneously. In such setups, activation maps from pre-trained 3D CNNs are quantified using statistical functions or Gaussian Mixture Models to generate compact distribution parameters, which are then concatenated with conventional radiomic vectors. This fused representation is fed into fully connected layers or survival analysis modules, such as Cox proportional hazards models, where all parameters are optimized jointly via gradient descent. This end-to-end training ensures that the deep feature extractor is implicitly guided to emphasize regions relevant to the handcrafted descriptors, effectively mitigating the noise associated with whole-image inputs.

Empirical evidence suggests that this joint optimization significantly improves performance in outcome prediction and segmentation tasks compared to isolated methodologies. By enforcing a shared latent space, the model reduces redundancy and maximizes the information gain from multimodal inputs, such as multi-parametric MRI or PET/CT scans. The approach not only alleviates overfitting in data-scarce scenarios by regularizing deep networks with prior knowledge encoded in handcrafted features but also discovers novel biomarkers that transcend human-defined constraints. Consequently, joint optimization represents a critical advancement in developing robust, clinically viable tools for precision oncology.

4.1.3 Genotype Guided Radiomics and Biological Priors

Genotype-guided radiomics represents a paradigm shift from purely phenotypic image analysis to the integration of underlying molecular architectures as biological priors [26]. By correlating high-dimensional imaging features with specific genomic alterations, such as IDH mutations in gliomas or MGMT methylation status, this approach decodes the radiographic phenotype to reflect intratumoral heterogeneity and evolutionary dynamics. This radiogenomic framework leverages non-invasive imaging to infer genetic profiles that traditionally require invasive biopsies, thereby facilitating a more comprehensive characterization of oncological diseases without the associated procedural risks [26].

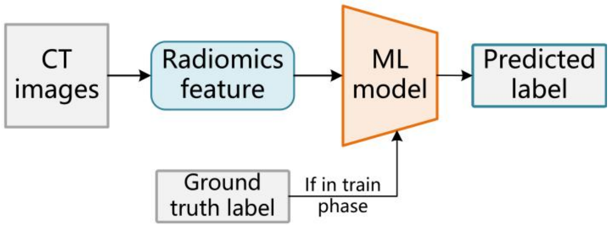

Figure 23: Chart from 'A Pilot Study of Relating MYCN-Gene Amplification with Neuroblastoma-Patient CT Scans' (arXiv:2205.10619)

The incorporation of biological priors addresses critical limitations in conventional radiomics, particularly regarding feature stability and interpretability [27]. Standard handcrafted features often suffer from sensitivity to segmentation variability and lack direct biological grounding. By constraining feature selection and model training with known genotype-phenotype associations, algorithms can prioritize imaging descriptors that are mechanistically linked to tumor biology. This guidance enhances the robustness of predictive models against inter-observer delineation errors and improves the generalizability of radiomic signatures across diverse patient cohorts and imaging protocols.

$$x^{(i,j)}\left(n^j\right) = \sum_{\sigma \in O} \frac{\exp\left(\alpha_\sigma^{(i,j)}\right)}{\sum_{\sigma' \in O} \exp\left(\alpha_{\sigma'}^{(i,j)}\right)} x_\sigma^{(i,j)}$$

Figure 24: Chart from 'Multi-Modality Information Fusion for Radiomics-based Neural Architecture Search' (arXiv:2007.06002)

Furthermore, merging genotype-informed radiomics with deep multi-task frameworks enables the joint learning of imaging patterns and clinical outcomes. Recent advancements utilize multiscale features from convolutional layers as learnable radiomic descriptors, which, when combined with genomic data, significantly improve survival prediction and treatment response assessment. This synergistic integration not only refines prognostic accuracy but also supports the development of adaptive therapy strategies by providing longitudinal, non-invasive biomarkers of tumor ecology, ultimately advancing the goals of precision oncology.

4.2 Deep Survival Modeling Frameworks

4.2.1 Neural Network Extensions of Cox Proportional Hazards

The Cox Proportional Hazards (CPH) model has long served as the gold standard for survival analysis, yet its inherent linearity restricts its ability to capture complex, non-linear interactions between high-dimensional covariates and event risk. To address this limitation, recent advancements have integrated deep neural networks with the CPH framework, replacing the traditional linear predictor with a non-linear function approximator. This architectural shift allows the model to learn intricate feature representations directly from raw data, such as medical images or genomic sequences, while retaining the semi-parametric nature of the original CPH formulation where the baseline hazard remains unspecified.

A prominent implementation of this paradigm is DeepSurv, which utilizes a deep feed-forward network to estimate the log-hazard ratio, optimizing parameters via the negative partial log-likelihood loss function. By mapping input covariates through multiple hidden layers with non-linear activation functions, these extensions effectively model heterogeneous treatment effects and personalized risk profiles that linear models often miss. Furthermore, variations such as Cox-Time introduce time-dependent covariates into the network architecture, enabling dynamic risk prediction that evolves over the follow-up period, thereby overcoming the static assumption of standard proportional hazards models.

These neural extensions have demonstrated superior predictive performance, particularly in domains characterized by high-dimensional, multimodal data where manual feature engineering is impractical. The integration of convolutional backbones for image analysis or recurrent layers for longitudinal data within the CPH loss framework facilitates end-to-end learning of prognostic signatures. Despite their enhanced capacity, these models require careful regularization to prevent overfitting on censored datasets and must maintain the proportional hazards assumption or explicitly model deviations to ensure clinical interpretability and robust generalization across diverse patient cohorts.

4.2.2 Discrete Time Hazard Models and Negative Log Likelihood

Discrete time hazard models address the limitations of continuous-time approaches by partitioning the survival timeline into distinct intervals, allowing for the estimation of conditional event probabilities within each bin. For a given subject, the hazard function h_j represents the probability of the event occurring in interval j, contingent upon survival up to the start of that period [28]. This discretization necessitates transforming continuous survival times into a sequence of binary outcomes, effectively recasting the survival problem as a series of logistic regression tasks across multiple time steps. By modeling the hazard directly rather than the cumulative distribution, these methods offer enhanced flexibility in capturing non-proportional hazards and time-varying covariate effects that traditional Cox models may overlook.

$$S_j^{(i)} = \prod_{q=1}^{j} (1 - h_q^{(i)}).$$

Figure 25: Chart from 'Enhanced Survival Prediction in Head and Neck Cancer Using Convolutional Block Attention and Multimodal Data Fusion' (arXiv:2410.21831)

The training of such models relies on minimizing the negative log-likelihood (NLL) of the observed data, which serves as the primary objective function. The likelihood formulation distinctly handles censored and uncensored observations: for subjects experiencing an event in interval j, the loss penalizes low predicted hazard rates, whereas for censored individuals or those surviving past interval j, it penalizes high hazard estimates. Mathematically, the total loss aggregates the negative log-probabilities across all relevant time intervals for every patient. The first term of the log-likelihood drives the model to increase the hazard score for events that actually occur, while the second term ensures the hazard remains low for intervals where the subject remains event-free, thereby accurately modeling the survival trajectory [28].

This framework facilitates the integration of deep learning architectures, where neural networks predict discrete hazard probabilities rather than a single risk score. Unlike the Cox partial likelihood, which relies on risk set rankings and assumes proportional hazards, the discrete NLL approach explicitly models the baseline hazard evolution over time through learned parameters for each interval. Consequently, this method provides a robust mechanism for estimating full survival distributions and handling complex, high-dimensional data modalities. The resulting probabilistic outputs enable precise calculation of survival functions and support more nuanced prognostic stratification compared to semi-parametric alternatives.

4.2.3 Deep Conditional Transformation Models for Distribution Estimation

Deep Conditional Transformation Models (DCTMs) represent a paradigm shift in survival analysis by directly modeling the conditional cumulative distribution function of event times rather than estimating hazard rates. Unlike traditional proportional hazards assumptions that constrain the relationship between covariates and risk, DCTMs learn a monotonic transformation function that maps the outcome variable to a predefined latent distribution, such as the standard normal. This architecture allows for the flexible approximation of arbitrary survival distributions, effectively capturing complex, non-linear interactions between high-dimensional medical imaging features and clinical variables without imposing restrictive parametric forms on the baseline hazard.

The core mechanism involves a deep neural network that parameterizes the transformation function, conditioned on input data derived from multimodal sources like radiomic features or voxel-wise image representations. By optimizing a likelihood-based loss function, the model ensures the learned transformation remains strictly monotonic, thereby guaranteeing valid probability estimates across the entire time horizon. This approach inherently handles censored data through the integration of the transformed distribution limits, enabling precise estimation of survival probabilities even when exact event times are unobserved. The flexibility of the deep backbone facilitates the extraction of prognostic representations directly from raw imaging data, bypassing the need for handcrafted feature engineering while maintaining statistical rigor.

Consequently, DCTMs offer superior performance in scenarios characterized by heterogeneous patient populations and complex disease trajectories where conventional regression models often fail. The ability to estimate the full conditional distribution provides clinicians with comprehensive risk profiles, including median survival times and confidence intervals, rather than single-point risk scores. This detailed distributional insight is critical for personalized treatment planning, as it allows for the evaluation of lifetime risks under varying clinical conditions. By bridging the gap between flexible deep learning architectures and rigorous statistical survival theory, these models establish a robust framework for accurate and interpretable prognosis in oncology and other time-to-event domains [29].

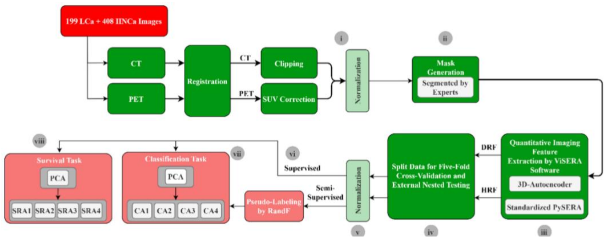

Figure 26: Chart from 'Enhanced Lung Cancer Survival Prediction using Semi-Supervised Pseudo-Labeling and Learning from Diverse PET/CT Datasets' (arXiv:2412.00068)

4.3 Multi-Task and Longitudinal Prediction Systems

4.3.1 Joint Segmentation and Survival Risk Stratification

Joint segmentation and survival risk stratification frameworks integrate tumor delineation directly with prognostic modeling to overcome the limitations of sequential pipelines. Unlike traditional radiomics that rely on manual or static segmentation, these end-to-end approaches utilize deep learning architectures, such as 3D nnU-Net or multi-task networks, to simultaneously optimize segmentation accuracy and survival prediction [7]. By sharing latent representations between the segmentation decoder and the survival regression head, the model ensures that the extracted radiomic features are inherently discriminative for time-to-event outcomes rather than merely maximizing geometric overlap metrics like the Dice Similarity Coefficient.

The core mechanism involves coupling segmentation masks with survival losses, often employing Cox Proportional Hazards or Multi-Task Logistic Regression objectives during training. This joint optimization allows the network to focus on specific tumor sub-regions, such as the necrotic core or enhancing rim, which carry distinct prognostic significance but might be overlooked by segmentation-only losses. Consequently, the predicted risk scores are derived from features that are spatially aligned with biologically relevant tumor phenotypes, enhancing the robustness of the model against segmentation errors and improving the concordance index by correctly ordering patient pairs based on their predicted hazard rates.

Empirical evaluations demonstrate that multimodal joint models consistently outperform baseline methods that treat segmentation and prognosis as disjoint tasks. These frameworks effectively stratify patients into high and low-risk groups, yielding Kaplan-Meier curves with statistically significant separation and higher hazard ratios compared to handcrafted signature-based approaches [30]. The integration of automatic segmentation eliminates observer variability, while the joint learning process ensures that the final survival predictions are grounded in precise anatomical boundaries, thereby facilitating more reliable clinical decision-making and personalized treatment planning.

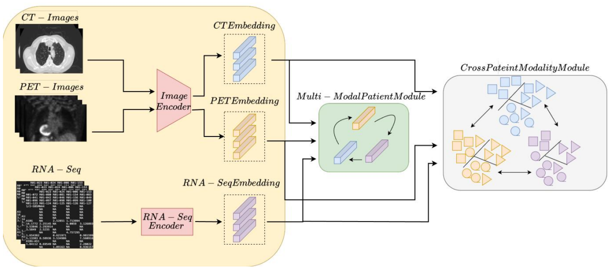

Figure 27: Chart from 'Survival Prediction in Lung Cancer through Multi-Modal Representation Learning' (arXiv:2409.20179)

4.3.2 Temporal Feature Integration via Recurrent Networks

Temporal feature integration via recurrent networks addresses the dynamic evolution of disease phenotypes by modeling longitudinal medical data as sequential inputs. Unlike static fusion methods that aggregate multimodal features at a single time point, recurrent architectures, particularly Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) networks, capture temporal dependencies inherent in serial imaging and clinical records. These mechanisms effectively process time-series data, such as weekly CBCT scans during radiotherapy or evolving electronic health record entries, to learn hidden states that represent the trajectory of tumor progression and treatment response over discrete intervals.

In the context of survival analysis, recurrent modules are frequently employed to encode spatio-temporal representations extracted from preceding convolutional backbones, such as 3D ResNets applied to volumetric PET/CT sequences. The recurrent layer aggregates these high-level deep radiomic features across time steps, allowing the model to weigh historical information against current observations dynamically. This sequential processing enables the network to identify critical inflection points in patient history that correlate with adverse outcomes, thereby refining the latent feature space before it is passed to downstream regression heads or Multi-Task Logistic Regression (MTLR) layers for hazard estimation.

Despite their efficacy in handling variable-length sequences and missing temporal data points, integrating recurrent networks into end-to-end survival frameworks presents challenges regarding computational complexity and interpretability. The recursive nature of these models can obscure the specific temporal contributions of distinct modalities, complicating the clinical validation of learned patterns. Consequently, recent advancements focus on hybrid architectures that combine attention mechanisms with recurrent units to explicitly highlight significant time steps and modalities, ensuring that the temporal integration not only improves predictive accuracy for overall survival but also provides actionable insights into the temporal dynamics driving prognosis.

4.3.3 Dynamic Radiomics and Time Dependent Covariate Modeling

Dynamic radiomics extends traditional static feature extraction by capturing temporal variations in tumor phenotypes throughout treatment trajectories [18]. Unlike conventional approaches that rely on a single baseline scan, this paradigm analyzes longitudinal imaging data to quantify intratumoral evolutionary dynamics and ecological shifts. By extracting time-series descriptors from sequential modalities, such as MRI or CT, researchers can monitor changes in texture, shape, and intensity distributions, thereby identifying early biomarkers of therapeutic response or resistance that static snapshots often miss.

To effectively leverage these temporal data streams, advanced statistical frameworks incorporate time-dependent covariates into survival analysis models. Traditional proportional hazards assumptions frequently fail to account for the non-linear and evolving influence of radiomic signatures on patient outcomes. Consequently, modern methodologies employ flexible non-parametric functions or discrete-time survival models that allow hazard ratios to vary over time. This approach enables the integration of dynamic imaging features alongside clinical variables, providing a more accurate representation of the complex, time-varying relationships between tumor morphology and mortality risk.

The implementation of dynamic radiomics often utilizes deep multi-task learning architectures capable of processing heterogeneous longitudinal inputs [15]. These frameworks aggregate deep features from convolutional neural network activation maps across multiple time points, fusing them with clinical metadata to predict survival probabilities at discrete intervals. By modeling the information flow through temporal layers, such systems overcome the limitations of static hand-crafted features, offering superior generalization and predictive performance. This shift toward dynamic modeling represents a critical advancement in personalized medicine, facilitating adaptive therapy strategies based on real-time phenotypic monitoring.

4.4 Interpretability and Clinical Validation

4.4.1 Attention Visualization and Saliency Map Generation

Attention visualization and saliency map generation have become indispensable tools for interpreting the decision-making processes of deep multimodal networks in medical imaging. By leveraging gradient-based backpropagation techniques, these methods produce heatmaps that highlight pixel-level regions contributing most significantly to model predictions, effectively bridging the gap between black-box algorithms and clinical interpretability. In the context of survival prognosis and tumor segmentation, such visualizations allow researchers to verify whether the network focuses on biologically relevant tumor subregions or spurious background artifacts, thereby validating the robustness of learned representations against noise and scanner variability.

Advanced architectures increasingly integrate explicit attention mechanisms, such as Convolutional Block Attention Modules (CBAM), to refine feature maps through sequential channel and spatial recalibration. Channel attention identifies "what" features are essential by re-weighting feature maps based on their interdependencies, while spatial attention determines "where" the network should concentrate, suppressing irrelevant contextual information. This dual mechanism enhances the discriminative power of fused PET/CT or MRI data, ensuring that the resulting saliency maps reflect a more precise alignment with pathological structures rather than diffuse activation patterns often observed in standard convolutional layers.

Despite their utility, current saliency approaches face challenges in systematically characterizing the histologic or radiomic features driving extreme risk scores. While entropy-based metrics and texture analysis offer compact descriptions of heterogeneity, they often fail to capture the full statistical range necessary for comprehensive explanation. Consequently, recent efforts focus on correlating saliency intensities with specific hand-crafted radiomic signatures, such as gray-level co-occurrence matrix entropy or kurtosis, to ensure that the visualized attention corresponds to quantifiable biological markers. This synergy between deep learning visualization and engineered features is critical for establishing trust in automated diagnostic systems and facilitating their adoption in clinical workflows.

4.4.2 SHAP Values and Feature Contribution Analysis

SHAP (Shapley Additive exPlanations) values serve as a critical mechanism for interpreting complex machine learning models by quantifying the marginal contribution of individual features to specific predictions [31]. In the context of survival analysis and radiogenomics, this game-theoretic approach assigns an importance value to each feature, such as Haralick textures, shape descriptors, or genomic markers, based on its impact on the model output relative to a baseline. By decomposing the prediction into additive feature contributions, SHAP facilitates a granular understanding of how diverse data modalities, including hand-crafted radiomics and deep learning-derived vectors, influence risk stratification outcomes.

$$Final\ Composite\ Score = \frac{1}{c \times n} \sum_{j=1}^{c} \left(\widehat{M}_j + Stability_j \right)$$

Figure 28: Chart from 'Robust Multicenter CT Radiogenomics for Dual EGFR and KRAS Prediction in Lung Cancer with Stability-Aware Modeling and SHAP Interpretation' (arXiv:2603.24922)

To ensure robustness and stability in feature importance assessment, SHAP values are often computed across multiple model configurations or cross-validation folds and subsequently aggregated [32]. This process involves calculating the mean absolute SHAP value for each prominent feature, which highlights those with the most consistent influence on classification boundaries or survival probabilities. Such aggregation mitigates the variance inherent in single-model interpretations, allowing researchers to distinguish between noise and genuine biological signals. Furthermore, separating these analyses by outcome classes, such as short-term versus long-term survival, reveals distinct feature interaction patterns that drive predictions for specific patient subgroups.

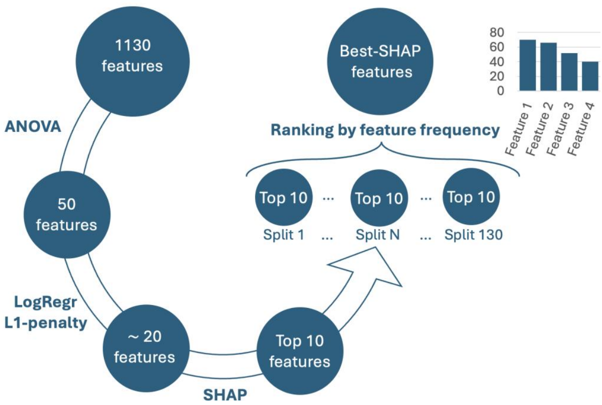

Figure 29: Chart from 'Segmentation variability and radiomics stability for predicting Triple-Negative Breast Cancer subtype using Magnetic Resonance Imaging' (arXiv:2504.01692)

The visualization of these contributions typically employs summary plots or heatmaps to illustrate the direction and magnitude of feature effects, thereby enhancing model transparency and clinical interpretability. For instance, positive SHAP values may indicate features that increase predicted risk, while negative values suggest protective factors, enabling a direct mapping between algorithmic decisions and biological plausibility. This interpretability is essential for validating that high-performing models rely on diagnostically relevant descriptors rather than artifacts, ultimately reinforcing confidence in deploying these systems for precision oncology and prognostic evaluation.

4.4.3 External Validation and Generalizability Assessment

External validation serves as the definitive benchmark for assessing the clinical utility and generalizability of prognostic models, moving beyond the optimistic estimates often derived from internal cross-validation. Rigorous studies employ completely independent external cohorts, distinct in terms of acquisition protocols, demographic distributions, and follow-up periods, to evaluate model robustness. This approach mitigates overfitting to specific institutional biases and ensures that performance metrics, such as the Concordance index and time-dependent AUC, reflect true predictive capability across diverse patient populations rather than memorization of training set idiosyncrasies.

To establish generalizability, researchers frequently adopt strict Type A validation strategies where models are retrained on the entirety of internal development data before being tested on unseen external sets. This process often involves aggregating results from multiple folds, where hyperparameters optimized via internal validation are fixed, and the final ensemble is applied to the external cohort without further tuning. Such methodologies highlight the disparity between internal and external performance, revealing that models relying solely on handcrafted features or single modalities often suffer significant degradation when faced with distributional shifts, whereas deep learning architectures integrating multimodal data demonstrate superior adaptability.

Furthermore, the assessment of generalizability extends to evaluating model stability under varying data volumes and through sensitivity analyses. By reporting performance exclusively on fixed external test sets while dynamically altering internal training splits, studies can isolate the impact of data scarcity on model convergence and feature selection consistency. Techniques such as SHAP value interpretation are subsequently utilized on these external cohorts to verify that the model relies on biologically plausible features rather than spurious correlations. Ultimately, consistent performance across independent datasets with differ-ing follow-up durations confirms the model's potential for broad clinical deployment and its resilience to temporal and spatial heterogeneity.

5 Future Directions

Despite significant advancements in integrating handcrafted radiomics with deep learning, several critical limitations persist that hinder widespread clinical adoption. Current methodologies often suffer from a lack of standardization in image acquisition and feature extraction, leading to poor reproducibility across different institutions and scanner types. While handcrafted features offer interpretability, they frequently fail to capture the full complexity of high-dimensional data without extensive manual engineering, whereas deep learning models, though powerful, operate as opaque black boxes that struggle with data scarcity and lack the transparency required for clinical trust. Furthermore, most existing studies remain retrospective and single-center, lacking the rigorous external validation necessary to prove generalizability in real-world heterogeneous patient populations. The fusion of topological data analysis with traditional pipelines also remains underexplored, leaving potential insights into global tumor connectivity and shape invariants untapped.

Future research must prioritize the development of robust, domain-adaptive frameworks that seamlessly merge the interpretability of classical radiomics with the representational power of deep neural networks. A promising direction involves creating hybrid architectures where handcrafted features, including those derived from wavelet transformations and persistent homology, serve as informed priors or attention mechanisms within deep learning models. This approach could mitigate the data hunger of neural networks while enhancing their ability to learn biologically meaningful patterns. Additionally, there is an urgent need for large-scale, multi-center prospective trials that adhere to strict standardization initiatives, such as the Image Biomarker Standardization Initiative, to validate these models across diverse imaging protocols. Research should also focus on advancing explainable artificial intelligence techniques specifically tailored for oncology, ensuring that model decisions can be traced back to specific textural or morphological biomarkers understandable by clinicians. Finally, exploring federated learning paradigms could enable the training of robust models on distributed datasets without compromising patient privacy, effectively addressing the challenge of limited sample sizes in rare cancer subtypes.

The successful execution of these proposed research directions holds the potential to fundamentally transform oncological care by delivering reliable, non-invasive biomarkers for precision medicine. By bridging the gap between algorithmic sophistication and clinical utility, future models will likely enable earlier detection of tumor progression, more accurate prediction of treatment responses, and personalized therapeutic stratification. Enhanced reproducibility and interpretability will foster greater confidence among healthcare providers, facilitating the integration of quantitative imaging into routine clinical workflows. Ultimately, these advancements promise to shift the paradigm from reactive treatment to proactive management, significantly improving patient survival outcomes and quality of life while reducing the burden of unnecessary invasive procedures.

6 Conclusion

This survey has systematically dissected the synergistic landscape of handcrafted radiomics and deep learning methodologies, elucidating their distinct yet complementary roles in oncological imaging. The review established that classical radiomics pipelines, grounded in sophisticated feature engineering such as statistical texture descriptors, wavelet transformations, and topological data analysis, offer unparalleled interpretability and robustness in data-scarce environments. These handcrafted features, when coupled with classical machine learning frameworks like Support Vector Machines and ensemble strategies including Random Forests and Gradient Boosting, provide reliable mechanisms for maximizing geometric margins and mitigating overfitting in high-dimensional spaces. Conversely, while deep learning architectures excel at automatic feature discovery, the analysis underscores that hybrid approaches fusing engineered and learned representations often yield the most clinically viable biomarkers. By critically evaluating feature selection techniques and dimensionality reduction strategies, this work clarifies the optimal deployment scenarios for each paradigm, demonstrating that the choice between manual engineering and automated extraction must be dictated by specific clinical objectives, data availability, and the requisite balance between predictive power and model transparency.

The significance of this comprehensive review lies in its holistic synthesis of theoretical algorithm development and practical clinical implementation, bridging a critical gap often left by previous studies that treat these domains in isolation. By explicitly mapping the interdependencies between feature engineering paradigms and classification frameworks, this survey provides a structured roadmap for researchers and clinicians to navigate the complex trade-offs inherent in medical image analysis. It serves as an essential reference for developing next-generation imaging biomarkers that are not only statistically robust but also reproducible and actionable within real-world healthcare settings. Furthermore, by highlighting the enduring value of interpretable models alongside emerging deep learning techniques, this work advocates for a balanced methodological approach that prioritizes clinical trust and biological plausibility, ultimately fostering the adoption of quantitative imaging as a standard component of precision oncology workflows.

Looking forward, the field must prioritize rigorous standardization and reproducibility to translate these advanced computational models from research prototypes into routine clinical practice. Future research directions should focus on harmonizing imaging protocols across institutions, validating biomarkers in large-scale multi-center cohorts, and integrating radiomic signatures with multi-omics data to capture the full spectrum of tumor biology. There is an urgent call for the community to adopt unified reporting standards and open-source frameworks that facilitate external validation and ensure that developed models generalize effectively across diverse patient populations. Only through such concerted efforts can the promise of data-driven decision support be fully realized, transforming medical imaging from a purely diagnostic tool into a dynamic predictor of treatment response and survival, thereby significantly improving patient outcomes in the era of precision medicine.

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